

## SUMMARY

Lead Data Scientist committed to creating human-centric AI solutions that drive organizational efficiency. Specialist in end-to-end product delivery, from data engineering to deployment, driving measurable KPI growth and strategic value within the Fintech and Retail sectors.

## EXPERIENCE

### [RappiCard Banorte.](#)

*Sr. Data Scientist → Lead Data Scientist*

México City, MX.

*(Sep 2024 – actual)*

- **Credit Line Increases and Primacy:** Orchestrated data-driven campaigns yielding **\$4M MXN incremental revenue** per semester. Leveraged Causal Inference (Double ML, ITS) to measure elasticity and design primacy strategies aimed at capturing market share and ensuring top-of-wallet competitiveness.
- **CLI Yield Optimization:** Architected and validated a Readiness Model for credit line increase (CLI) campaigns; defined the targeting strategy that projected a 53% (\$8M MXN) profit growth, currently serving as the blueprint for upcoming portfolio expansions.
- **Profitability Segmentation:** Developed **unsupervised ML models** on **AWS SageMaker** to create high-stability behavioral segments; achieved consistent cluster assignments to drive long-term profitability strategies and integrated a **GenAI agent** to automate stakeholder reporting and insight generation.
- **RC Score (Risk Forecasting):** Engineered a **stacked LSTM + XGBoost ensemble** to process sequential transactional data (**0.90 AUC**). Architected a production pipeline on **AWS SageMaker** (S3, Training Jobs) via **Docker and Airflow**, serving as the core risk engine that maintained the **Índice de Morosidad (IMOR) below 5%** during aggressive portfolio-wide CLI campaigns.
- **LTV Modeling:** Developed an advanced Lifetime Value forecasting model, improving accuracy by 10%; provided the analytical foundation for strategic finance decisions, capital provisioning, and CAC optimization.
- **Leadership:** Mentored 2 Junior Data Scientists; took technical ownership of portfolio-wide risk and growth initiatives, reporting directly to the CEO and Executive suite.

### [PhiQus Data Science /ENKI.](#)

*Data Scientist Consultant*

México City, MX.

*(Oct 2023 – May 2024)*

- **Contributed to the founding of PhiQus, formed by the Data Science team at the company I previously worked with (ENKI)**
- **Logistics Optimization (Soriana):** Designed and deployed a route optimization algorithm via Azure Functions, significantly reducing delivery times and mileage for online orders.
- **Operational Optimization:** Boosted Soriana's inventory "Found Rate" by engineering a production ML pipeline for **10k+ products**; developed high-impact features that reduced **RMSE by 20%** to drive in-app product availability decisions.
- **Causal Analysis:** Used **Difference-in-Differences** to quantify a **12% sales uplift** for Soriana, cutting inefficient marketing spend by 25%.
- **Client Acquisition:** Successfully onboarded **Home Depot MX** by designing a high-performance **ARIMA + ML PoC** that outpaced baseline forecasts and secured executive-level buy-in.
- **Standardization:** Engineered the company's first DS Documentation Framework, improving cross-team reproducibility and code quality.

## EDUCATION

**M. Sc. in Data Science:** Instituto Tecnológico Autónomo de México (ITAM)

**BA. Sc. in Applied Mathematics:** Universidad Autónoma del Estado de Hidalgo (UAEH)

**Diplomas:** Applied Deep Reinforcement Learning

GPA 9.6/10. (Full Excellence Scholarship)

GPA 9.59/10. (Top Graduate / Mérito Académico)

(Certification available [here.](#))

## TECHNICAL SKILLS

- **Cloud and big data:** AWS (SageMaker, S3, Lambda, Step Functions, Athena, EC2, Glue), Azure (Functions and Azure DevOps), Snowflake.
- **Data Science:** Python, R Software, SQL.
- **AI frameworks:** ScikitLearn, econML, XGBoost, Hugging Face, PyTorch.
- **Business Intelligence:** Shiny and Streamlit.
- **Bayesian Inference:** Stan.

## SELECTED PROJECTS & CERTIFICATIONS

**PERSONAL AI LAB:** Fine-tuned **LLaMA** via LoRA/RLHF for deceptive text detection (*Repo available [here.](#)*)

Built an **Actor-Critic** movie recommendation engine (Offline RL) using PyTorch. (*Repo available [here.](#)*)

**CERTIFICATIONS:** Machine Learning in Production – **DeepLearning.ai** (Certification available [here](#))

Spark and Datalakes – **Udacity** (Certification available [here.](#))

Agentic AI Nanodegree – **Udacity** (On going)